STRUCTURAL SPECIFICATION (SEED DRAFT)

PJR-2026-003_WAREHOUSE (STEAM BOILER)

Document control

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Project	PJR-2026-003 WAREHOUSE (STEAM BOILER)
Client	Mr. XXX
Stage	FOR APPROVAL
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Design basis	PJR-2026-003 design basis, status DRAFT
National Annex	CEN_RECOMMENDED (Cambodia)
Prepared by	P.LIHENG
Checked by	
Approved by	

Seed draft generated from the issued WH-STR-SD R00 notes and the design basis. Specification standard 07 is not yet authored; format is PROPOSED.

1 GENERAL REQUIREMENTS

1.1 The structure shall be designed to the Eurocodes listed below using CEN recommended parameter values; local authority requirements shall apply where more onerous. *[DB: national_annex; SD-003 B]*

Design codes (from design basis)

Standard	Generation	Edition	Used for
EN 1990	NOT RECORDED	NOT RECORDED	Basis of structural design
EN 1991-1-1	NOT RECORDED	NOT RECORDED	Self-weight, imposed loads
EN 1991-1-3	NOT RECORDED	NOT RECORDED	Snow
EN 1991-1-4	NOT RECORDED	NOT RECORDED	Wind
EN 1992-1-1	NOT RECORDED	NOT RECORDED	Concrete structures (RC)
EN 1993-1-1	NOT RECORDED	NOT RECORDED	Steel - general rules
EN 1993-1-3	NOT RECORDED	NOT RECORDED	Cold-formed members (purlins/girts)
EN 1993-1-5	NOT RECORDED	NOT RECORDED	Plated structural elements
EN 1993-1-8	NOT RECORDED	NOT RECORDED	Joints
EN 1997-1	NOT RECORDED	NOT RECORDED	Geotechnical design

Mixed 1st/2nd generation Eurocodes: User decision (P.Liheng, 14-OCT-2026): 1st and 2nd generation Eurocode parts may be combined on this project. The generation of each part must still be recorded per standard entry; compatibility of each combination is for the engineer to confirm.

National Annex: CEN_RECOMMENDED. Per SD-003: CEN recommended values used (no National Annex for Cambodia); local authority requirements apply where more onerous. Confirmed by user (P.Liheng) 14-OCT-2026.

1.2 Concrete work shall be executed to EN 13670. *[SD-004 notes]* **[SEED - verify]**

1.3 Steelwork fabrication and erection shall be to EN 1090-2, execution class EXC2. *[SD-005 C]* **[SEED - verify]**

2 MATERIALS

Materials (from design basis)

Id	Element	Specification	Status	Source
M-CONC-STUMP	Stump columns / stiffener	C30/37 XC1/XC4	confirmed	SD-004 A

Id	Element	Specification	Status	Source
	columns			
M-CONC-PILE	Pile caps / tie beams / ground beams	C30/37 XC2	confirmed	SD-004 A
M-REBAR	Reinforcement	fyk = 390 MPa deformed bars, grade 390	confirmed	SD-004 A
M-STEEL-MAIN	Plates / welded sections / hot-rolled	S355JR (EN 10025-2)	confirmed	SD-005 A
M-BOLTS	Structural bolts	Class 8.8 / 10.9 (EN 14399-3 HRPV preloaded where noted)	confirmed	SD-005 A

2.1 Reinforcement shall be deformed bars with characteristic yield strength fyk = 390 MPa. *[DB: M-REBAR]*

2.2 Structural steel plates and welded sections shall be S355JR to EN 10025-2. *[DB: M-STEEL-MAIN]*

3 WORKMANSHIP

3.1 Concrete shall be compacted with poker vibrators and wet cured for a minimum of 7 days (EN 13670 curing class 2). *[SD-004 E]* **[SEED - verify]**

3.2 No reinforcing bar shall be rebent on site except as permitted by EN 13670. *[SD-004 D]* **[SEED - verify]**

3.3 No steel erection shall start until the stump columns have achieved 70 % of fck and the anchor bolts have been surveyed. *[SD-004 E]* **[SEED - verify]**

4 QUALITY CONTROL

4.1 Concrete compression testing shall be one set of three cubes per pour or per 50 m3, tested at 7 and 28 days to EN 12390. *[SD-004 E]* **[SEED - verify]**

5 ASSUMPTIONS TO BE CONFIRMED

Id	Open item (from design basis)
F1	Foundation - precast square driven piles 300x300, L=7.0 m; soil investigation and pile load tests required
F2	Boiler / piping / crane / hoist and any platform loads not included
F3	Openings (doors, louvres, flue) not shown
F4	Terrain category II and no shielding, site altitude < 100 m
F5	Longitudinal (grid A/B) wind sway by braced bays - confirm
F6	Purlins/girts cold-formed - supplier verification for loads in SD-108
F7	Client name is a placeholder
F8	Eave sway limit h/100 = 120 mm - confirm with cladding supplier